

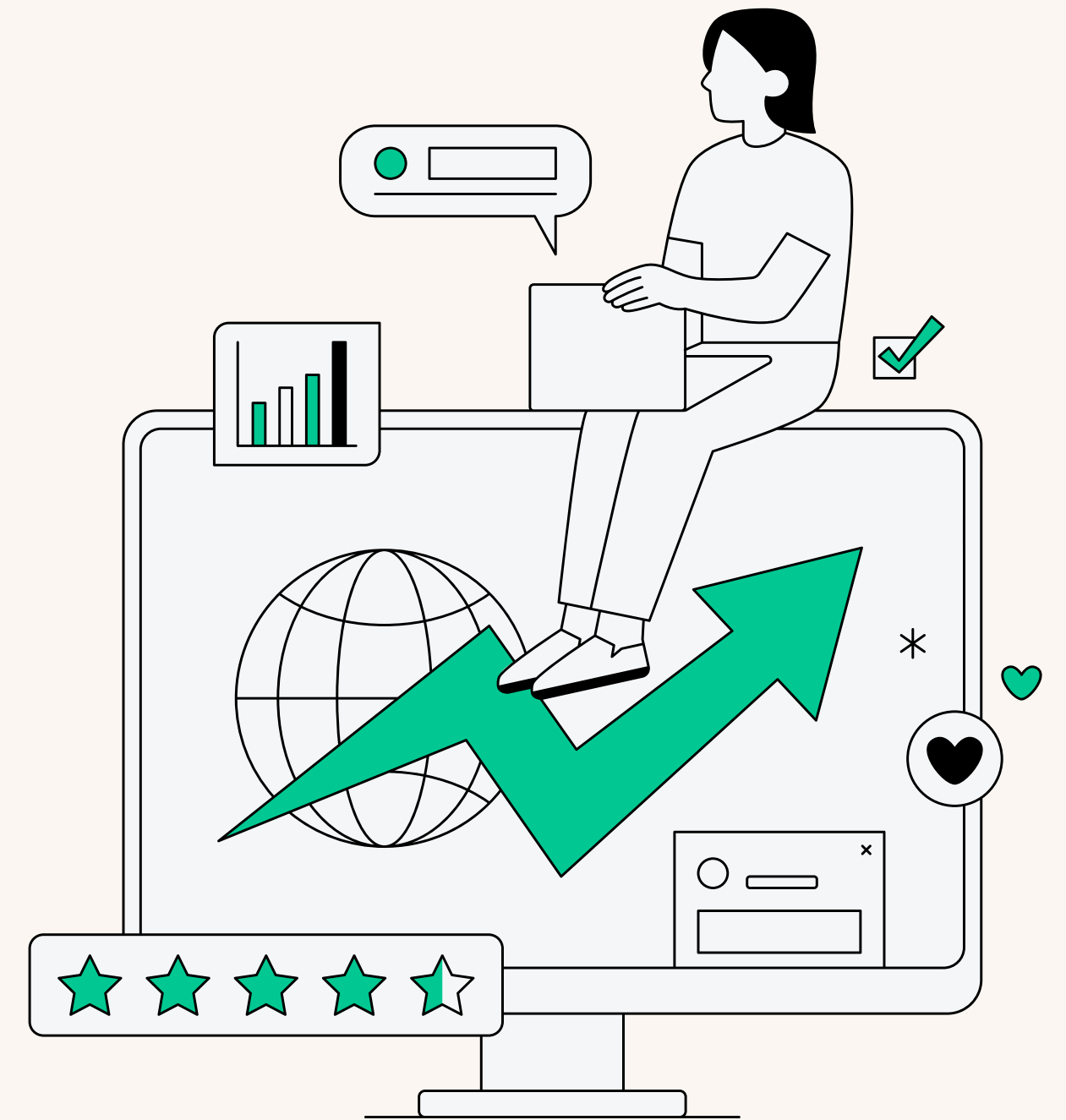
Stratify

Business Analytics Bootcamp - 2025

Project 2 – QuickBasket (Decoding the Basket)

Endterm Report

Presented by – Shubh Sareen and Tanmay Sonkar

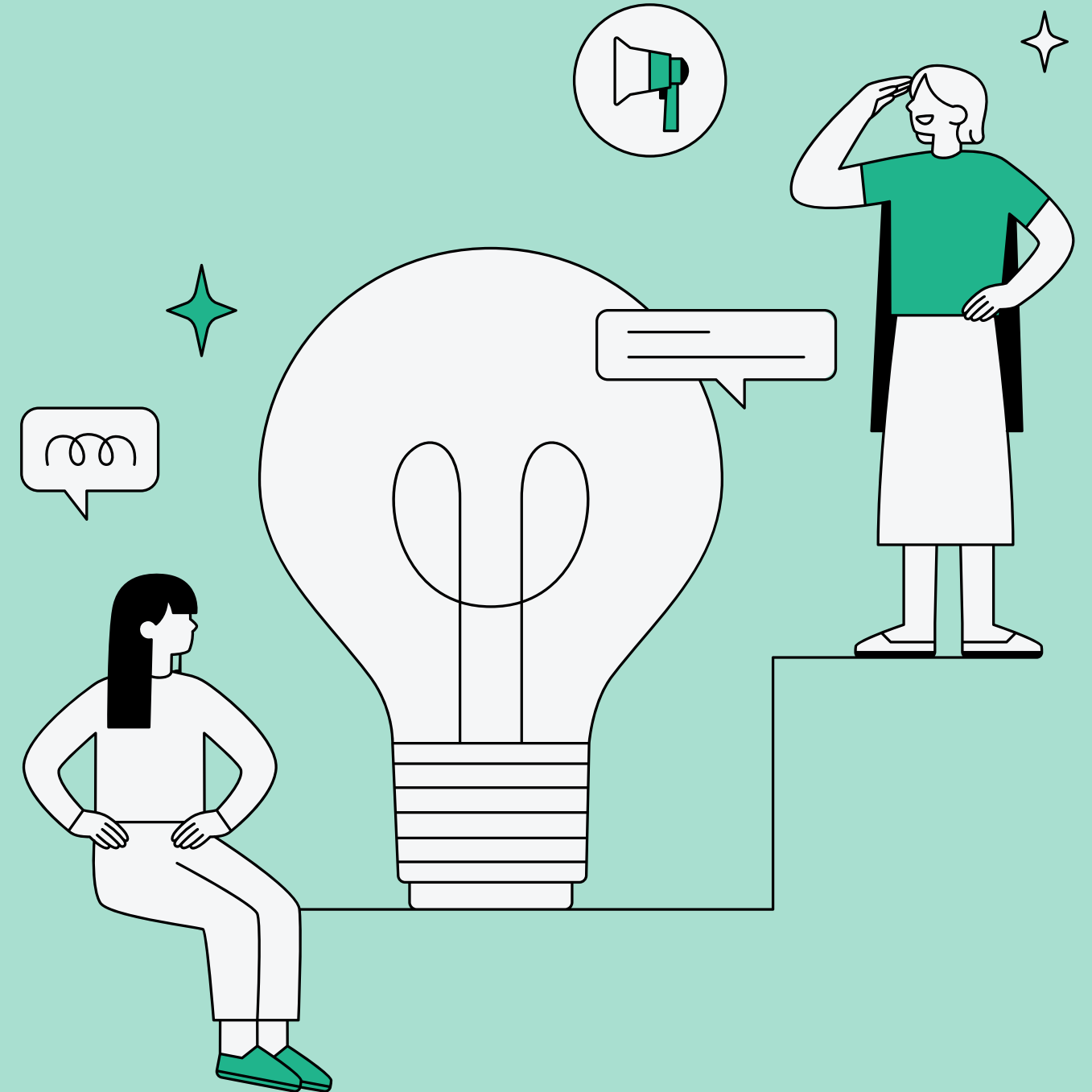


Basics of Data Analysis

Task 1 :

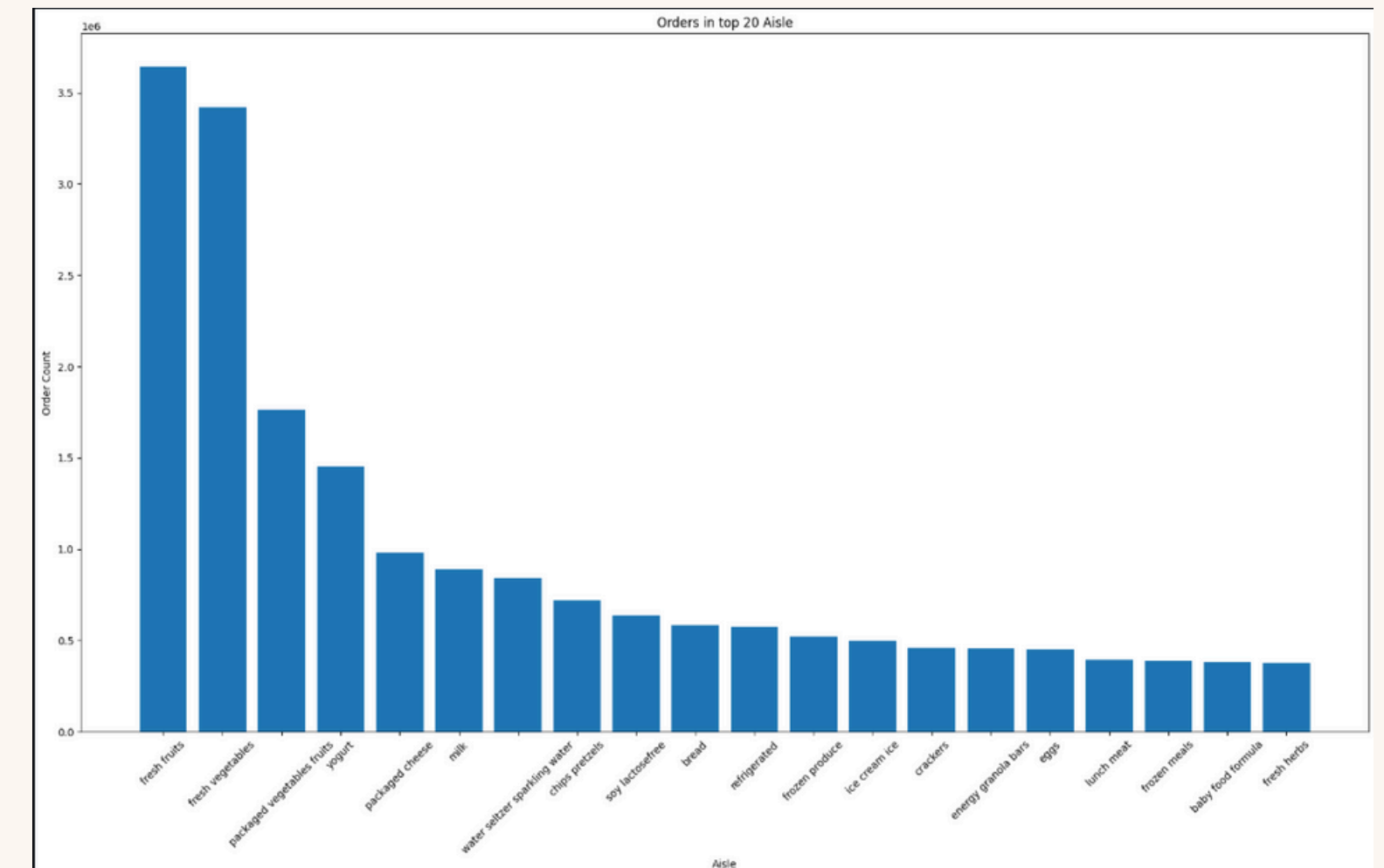
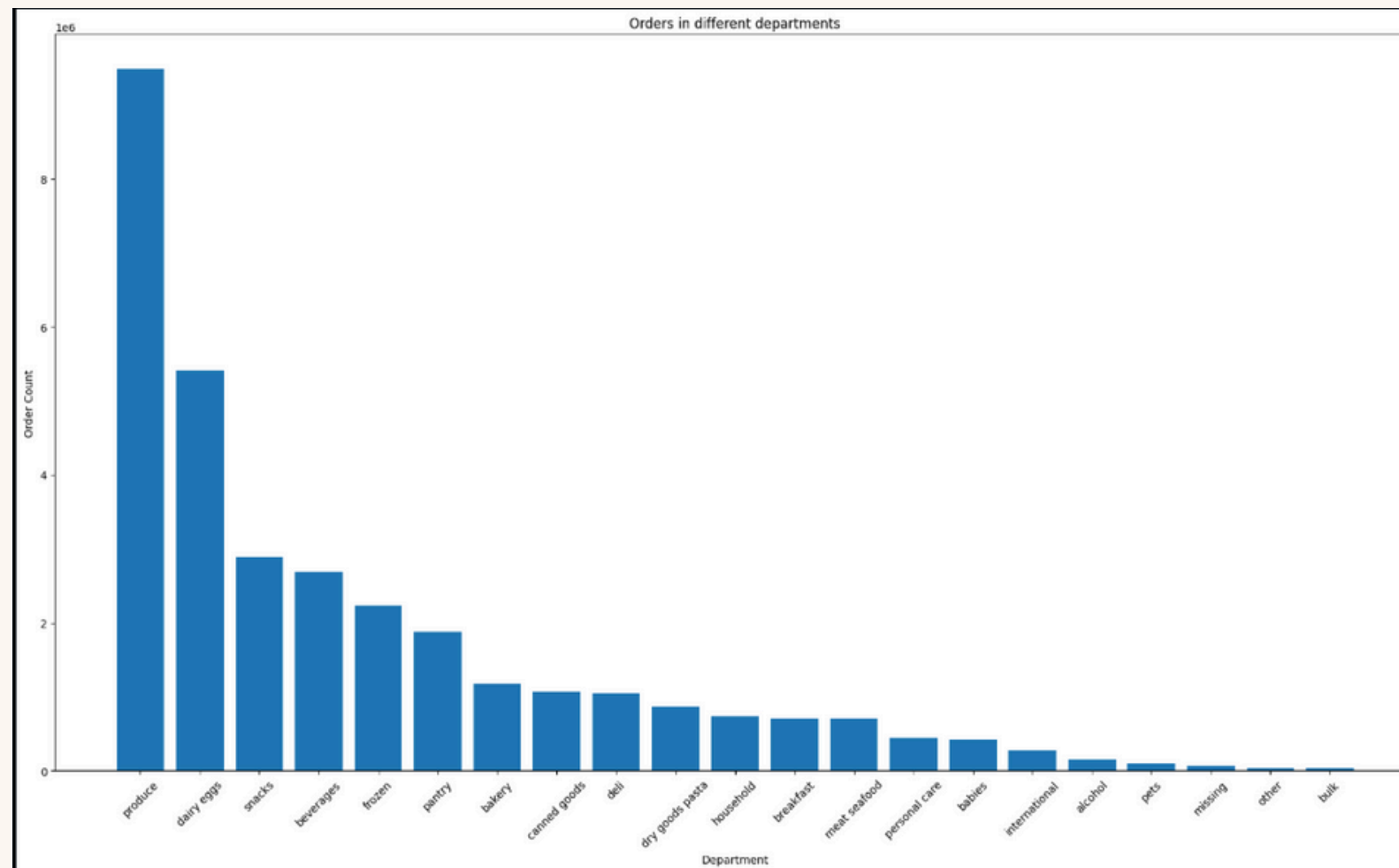
Exploratory Data Analysis

Using Pandas and Matplotlib



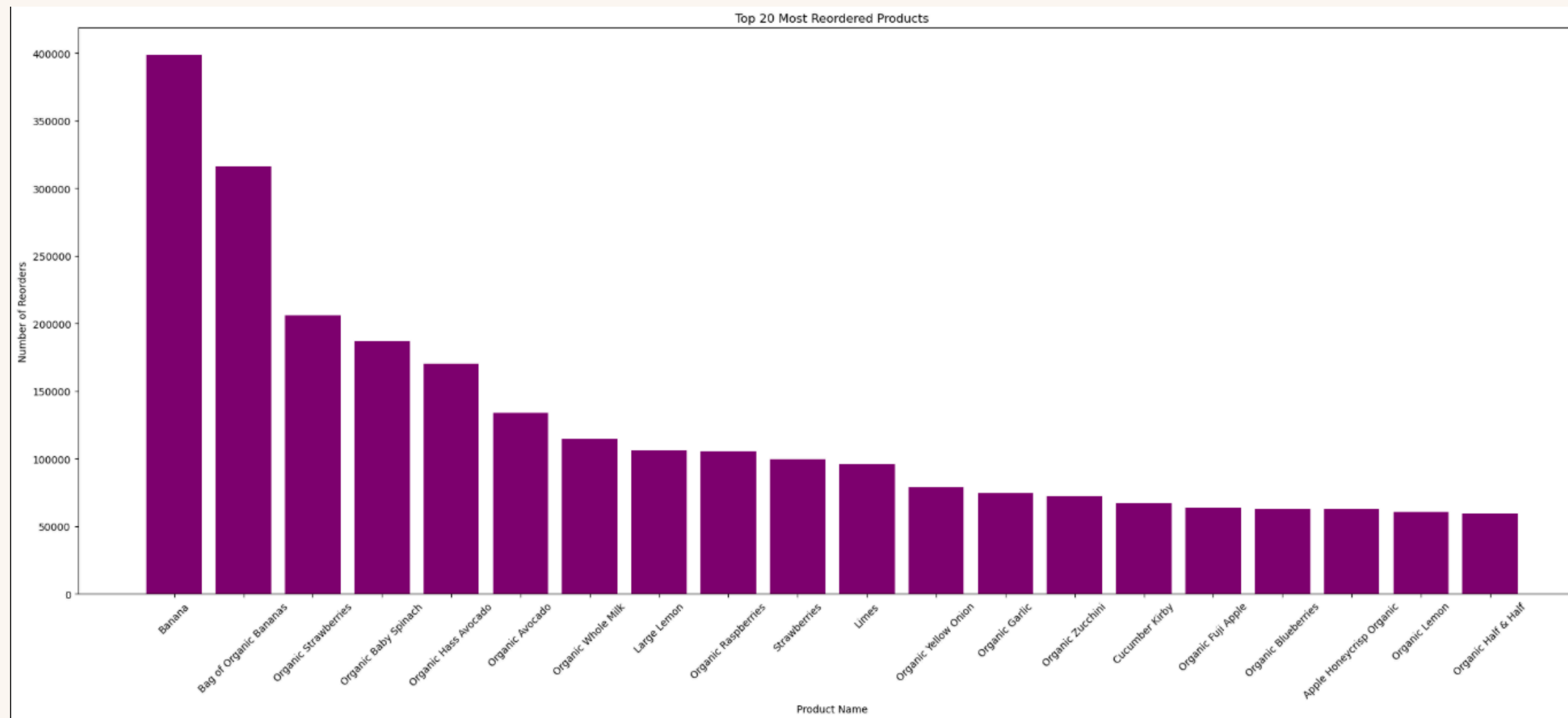
Data Grouping & Ordering

- We can group datasets by categories (e.g., departments, aisles) and count order volumes.
- This helps identify which categories drive sales.
- Visualization: Bar charts of top departments & aisles.



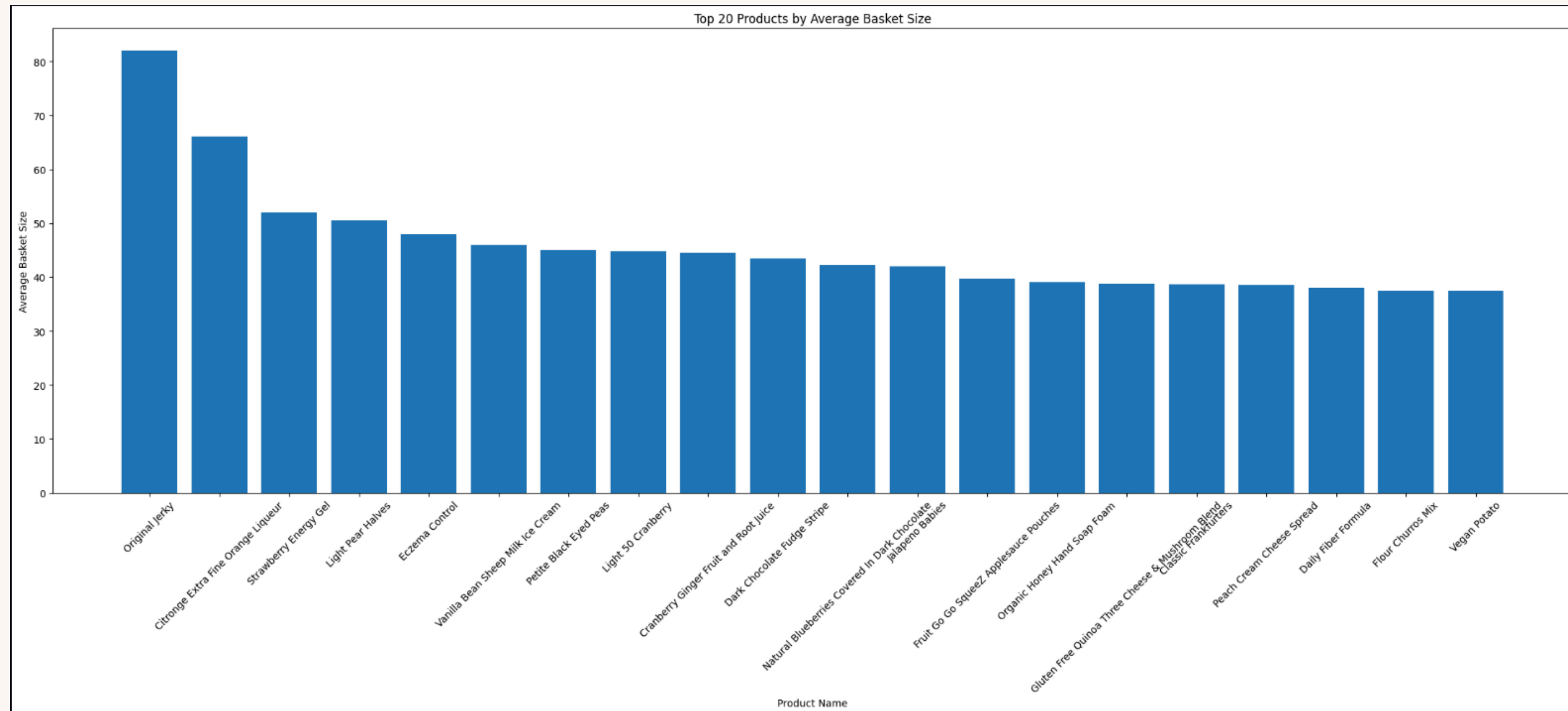
Reorder Rate Analysis

- Reorder rate = fraction of times a product was reordered after first purchase.
- This helps identify products with high loyalty.
- Visualization: Table/Chart of products with highest reorder rates.



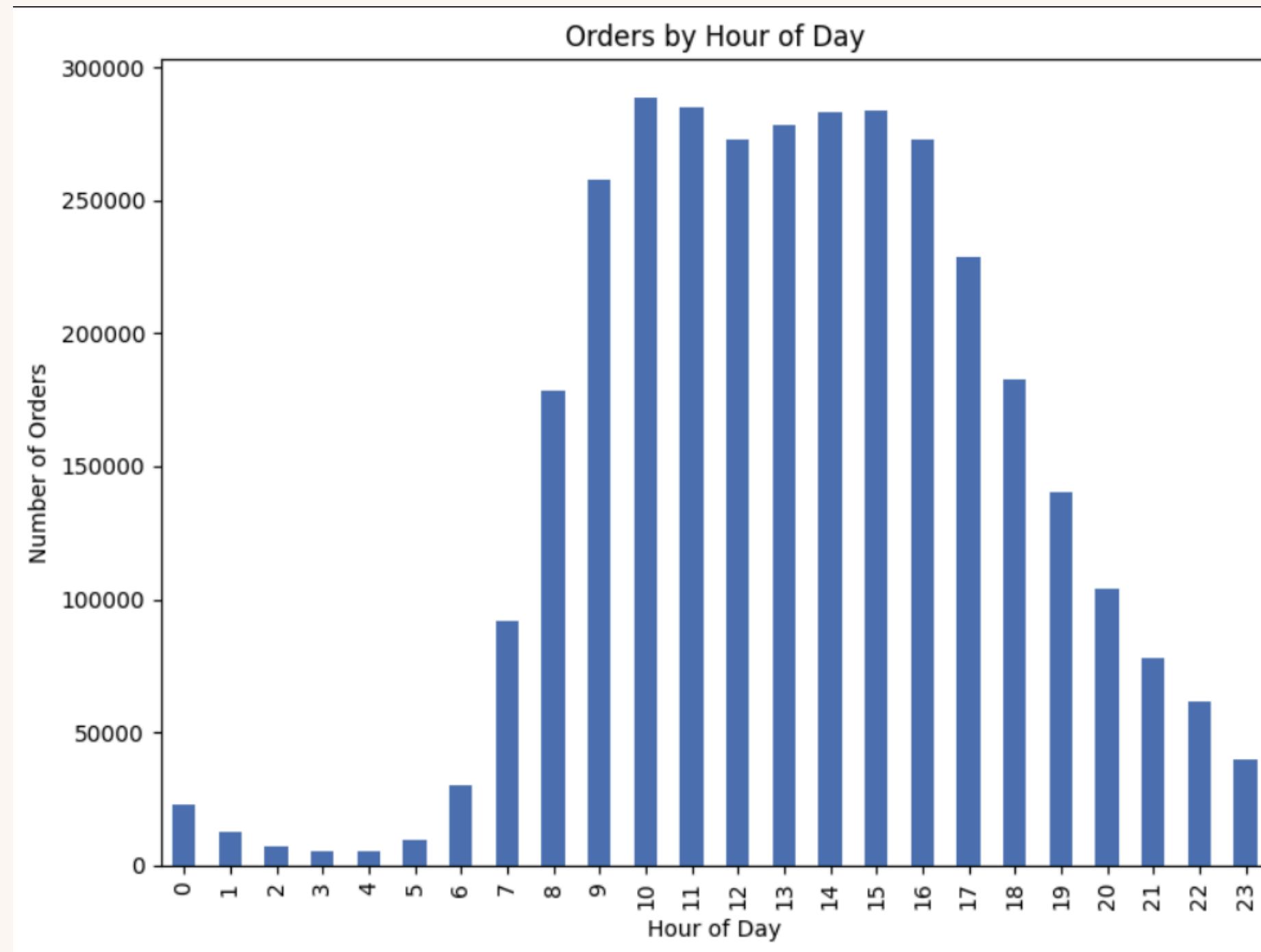
Basket Size Distribution

- Basket size = number of items in a single order.
- We can calculate average and median basket sizes, and visualize the distribution.
- Visualization: Histogram of basket sizes.



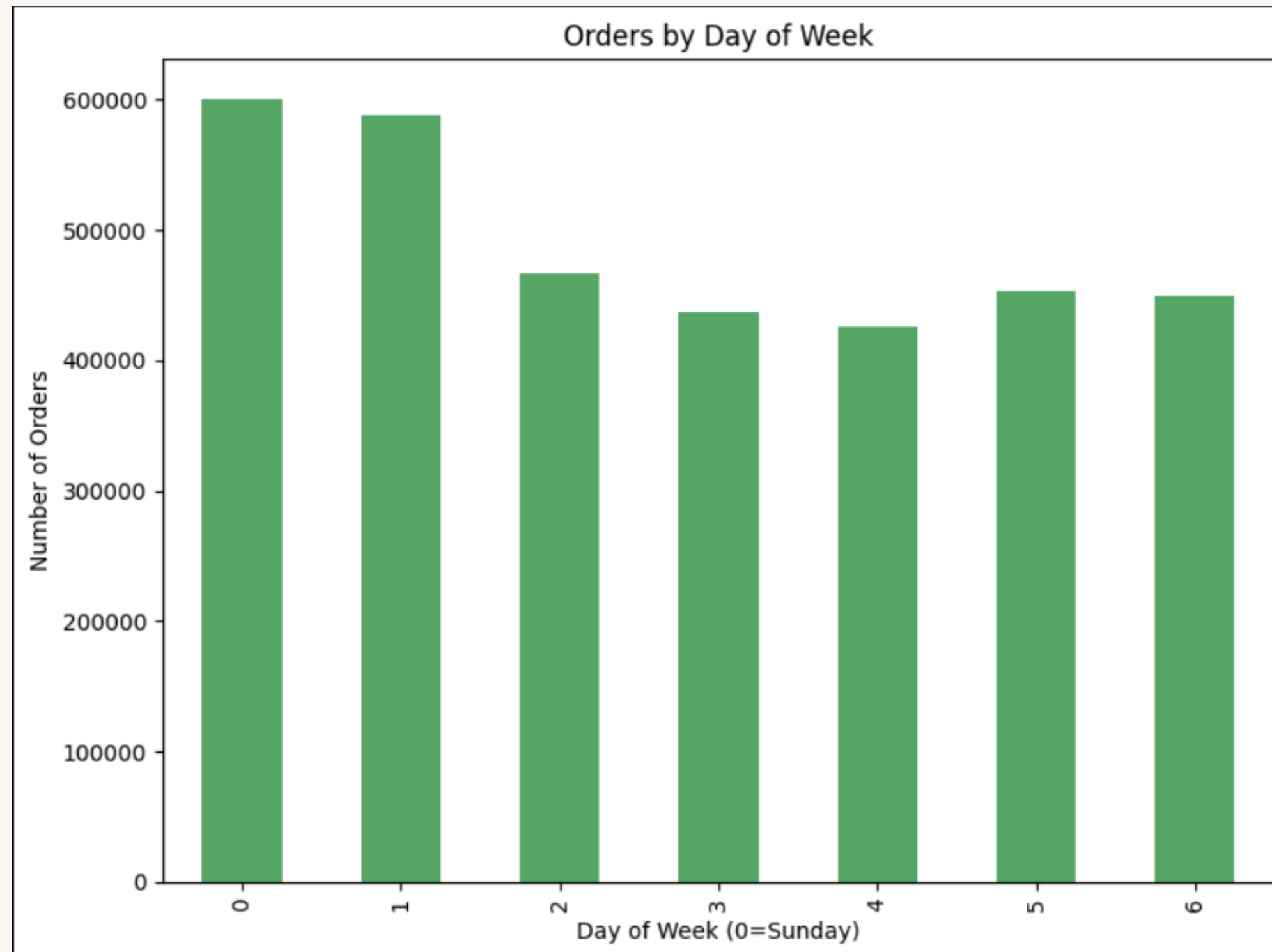
Order Trends by Hour of Day

Analyzing order timestamps helps us understand customer activity patterns.



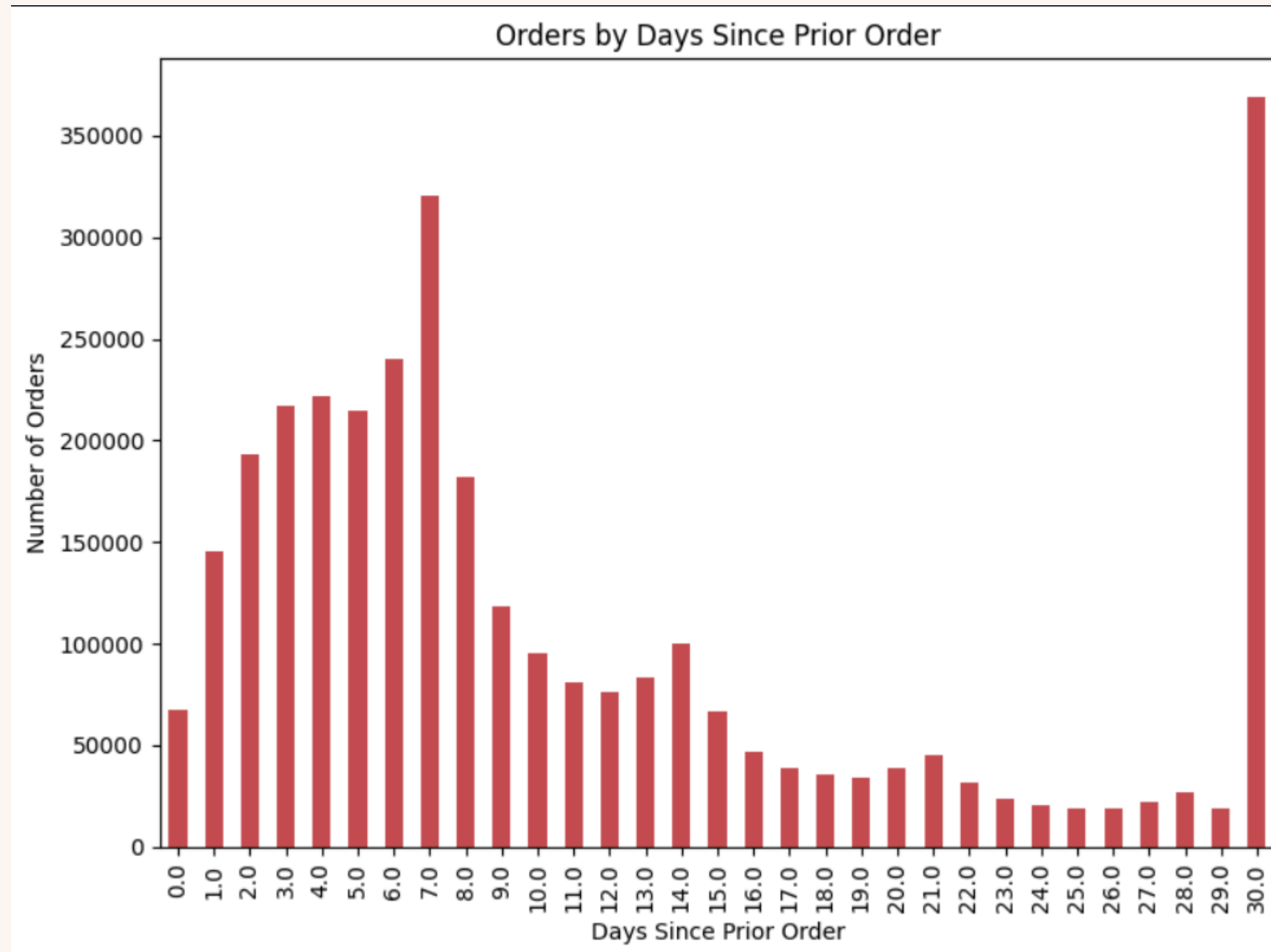
Order Trends by Day of Week

Customers may shop differently on weekdays vs weekends.



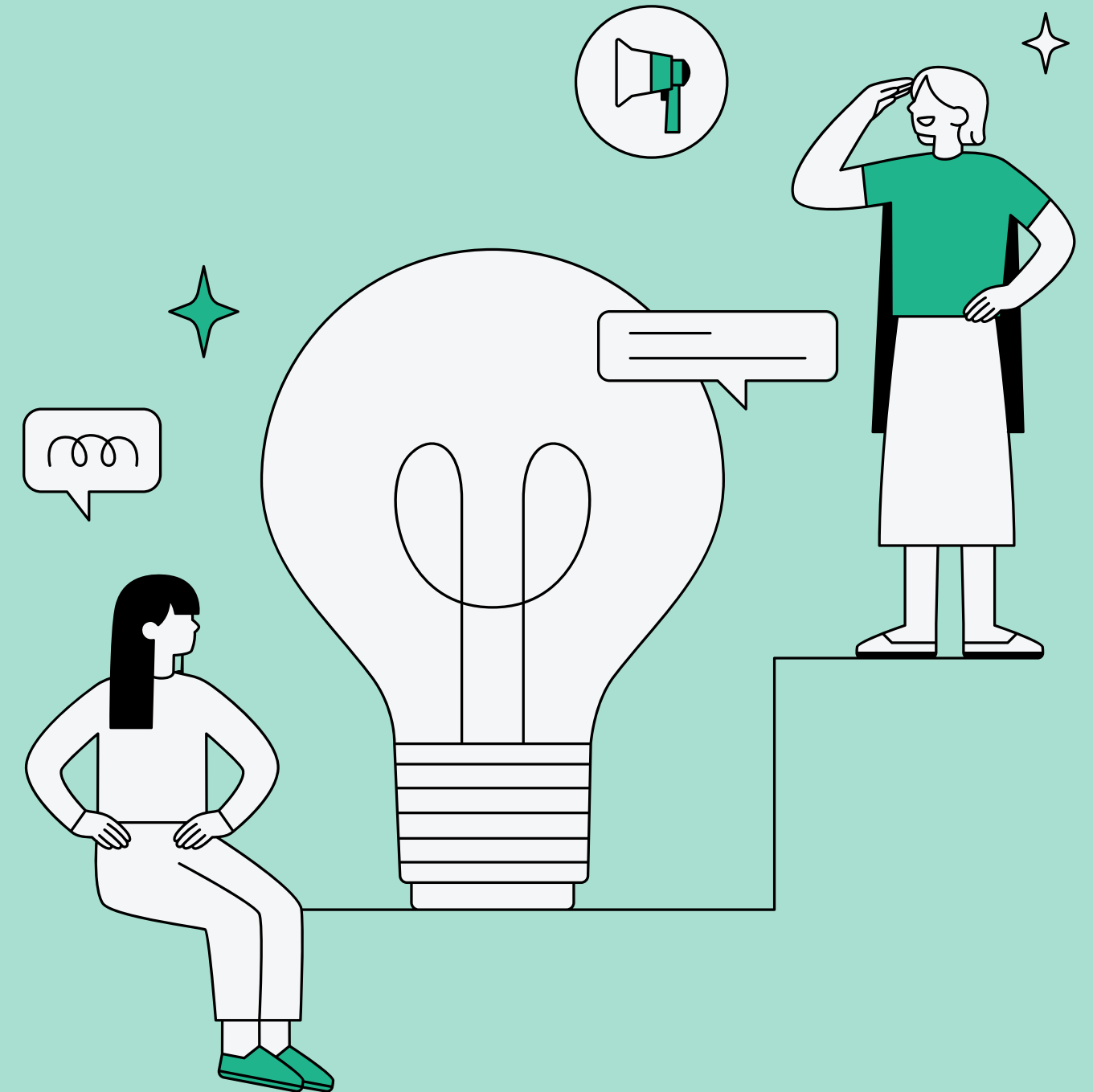
Days Since Prior Order

This metric shows repeat purchase intervals.



Task 2 :

Understanding Customer Buying Patterns using Product Combination Analysis



Popular Combinations

Bag of Organic Bananas → Organic Hass Avocado
Bag of Organic Bananas → Organic Strawberries
Banana → Organic Strawberries

Top Product Pairs

- **Frequent Pairs:** example, “Bag of Organic Bananas + Organic Hass Avocado” appears 64,761 times (2.65% of all transactions).
- Bananas, avocados, and strawberries are common across many pairs → indicating they are basket anchors.
- **Customer Behavior Insights :** Customers often buy fruits and vegetables in bundles (bananas + strawberries, bananas + spinach).
- Suggests health-conscious or fresh produce–focused baskets.

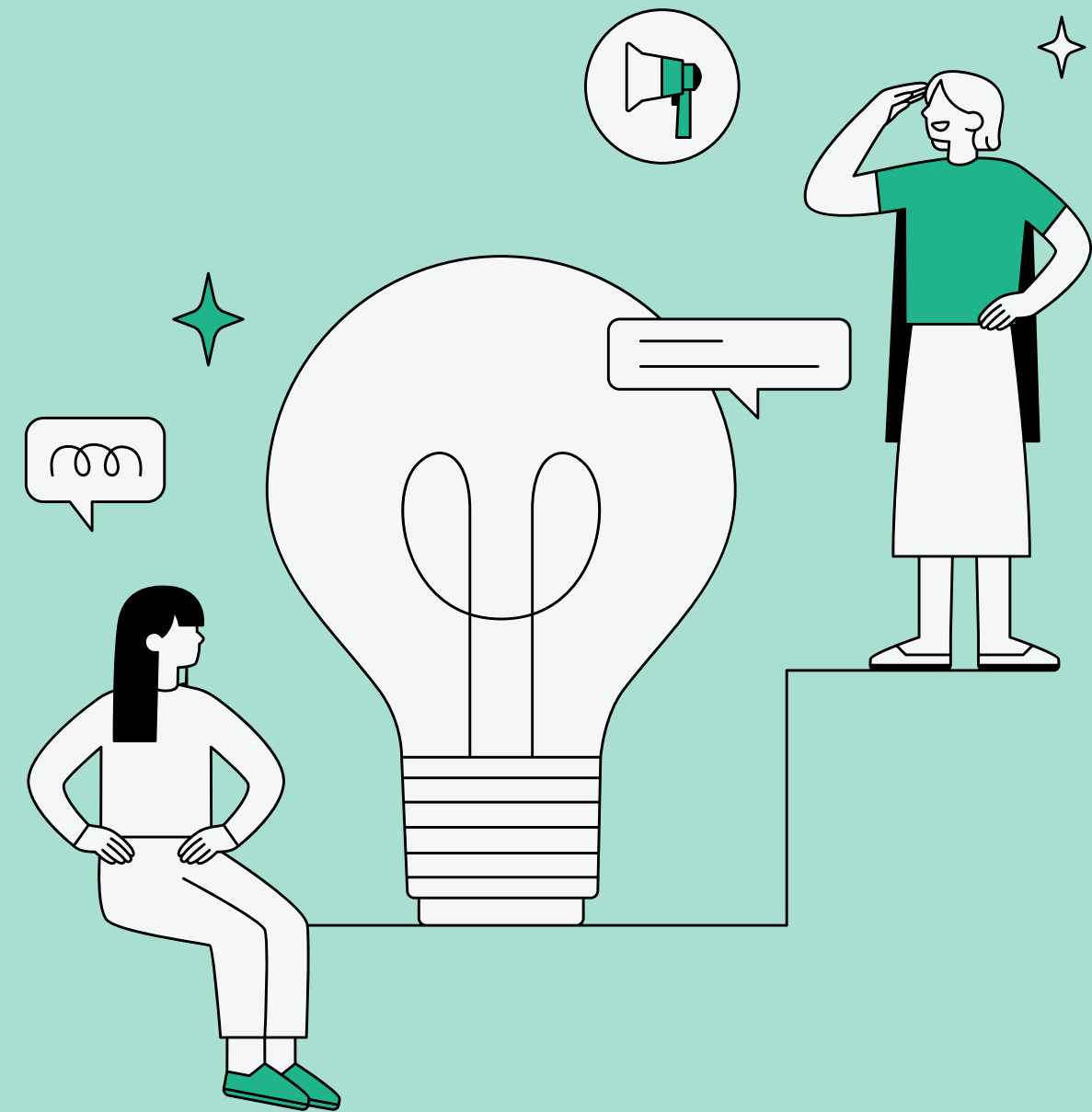
Top Product Pairs

- **Cross-Selling Opportunities:** High-frequency pairs mean if a customer buys product A, product B has strong upsell potential.
- Example: If someone buys Organic Bananas, recommend Organic Strawberries or Avocados.
- **Product Affinity Strength:** The support metric helps rank which pairs are most influential across all sales.
- Even though 2–3% seems small, in retail data this is very significant.

Task 3 :

Business Recommendations

Based on inferences collected
from the data



Inventory Suggestions

- Frequent Replenishments and higher safety stock for top seller items like bananas, strawberries and avocados.
- Set dynamic safety stock proportional to weekly sales velocity, reorder frequency, and supplier lead time. For fastest movers, target safety stock = $2-4 \times$ lead-time demand.

Product bundling

- Bundle frequently bought products together like Bananas with Avocados, Strawberries or Spinach.
- Place frequently bought products close to each other in the aisle.
- Give discounts on bundles to increase sales.

Time for sending promotional notifications

- Key windows: During peak hours (10am to 4pm) and early morning for daily groceries.
- Send a first wave 1-2 hours before the highest morning peak and a second wave 30-60 minutes before the main evening peak.

Time for sending promotional notifications

- Weekly notifications can be sent on Friday and Saturdays, catering to the Sunday peaks every week.
- Similarly Monthly notifications can be sent after 28 days of last purchase, catering to the large peak at day 30.

Recommendations for featured products in the app homepage / banners

- Rotate banners weekly and A/B test combinations: single best-seller banner vs. “Bundle of the Week”.
- Track KPIs (Key Performance Indicators) for each featured tile (impressions → CTR → add-to-cart → conversion) and reallocate prime homepage space to tiles that show positive lift within 7–14 days.

Thank You !!